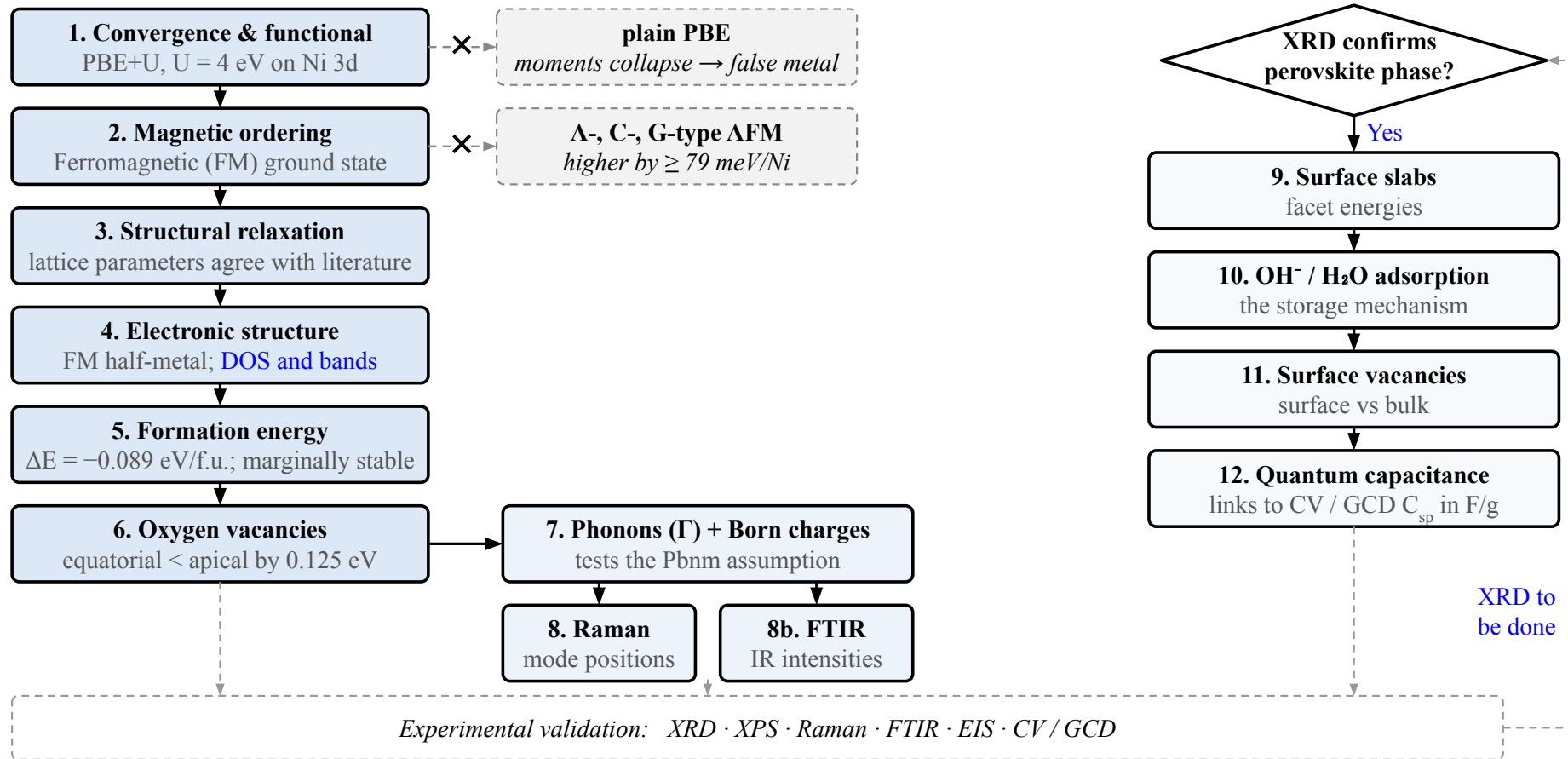
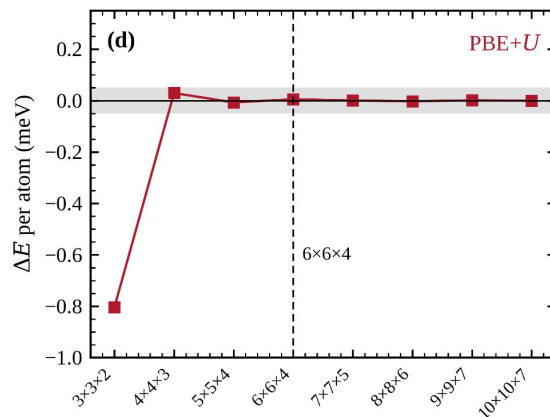
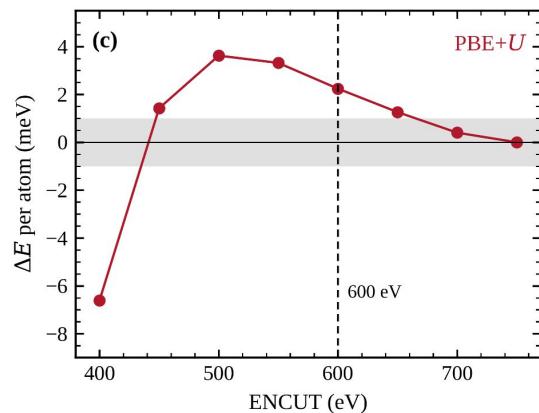
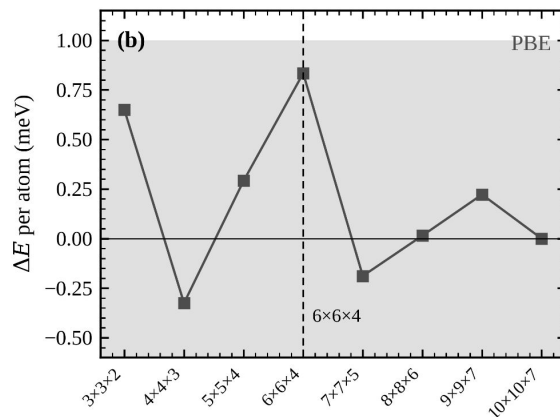
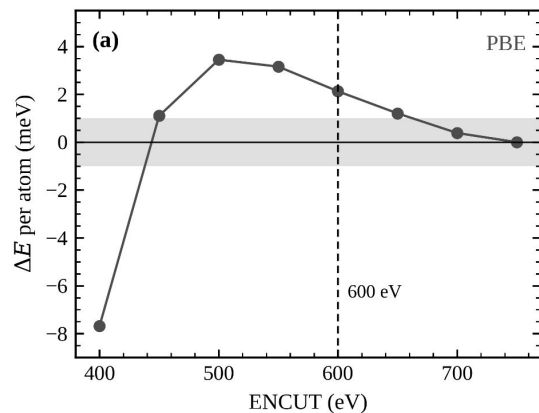


Density Functional Theory on GdNiO_3

DFT workflow: GdNiO₃ supercapacitor electrode



Convergence testing & Functional choice (plain PBE and PBE+U)



Plain PBE collapsed the Ni moments to $\pm 0.005 \mu_B$, producing a non-magnetic metal with $11.26 \text{ states eV}^{-1}$ at the Fermi level; inconsistent with the known magnetically ordered, insulating character of GdNiO_3 .

Convergence testing: plain PBE and PBE+U

Test	PBE	PBE+ <i>U</i>	Adopted
<i>k</i> -point mesh	Scatter ± 0.8 meV/atom	< 0.04 meV/atom above $4 \times 4 \times 3$	$6 \times 6 \times 4$
ENCUT	Non-monotonic, ~1 meV/atom residual	Non-monotonic, ~2 meV/atom residual	600 eV

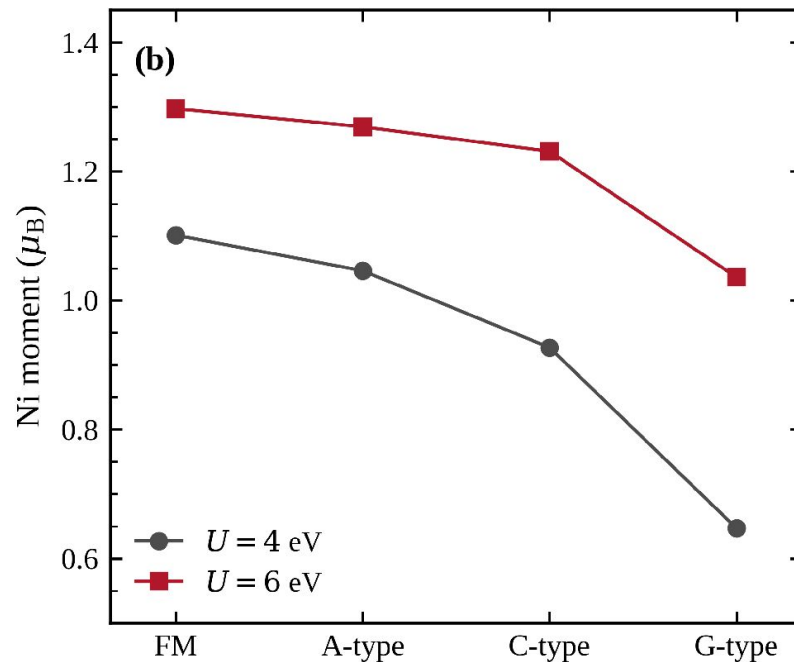
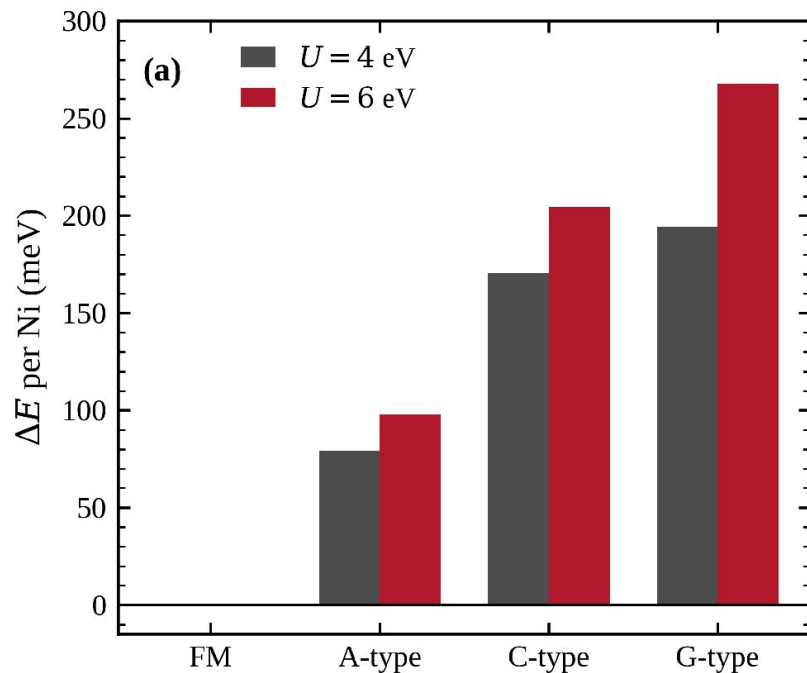
Magnetic ordering at $U = 4$ and $U = 6$ eV

Four configurations: ferromagnetic (FM) and A-, C- and G-type antiferromagnetic were compared at fixed experimental geometry, and the test repeated at $U_{\text{eff}} = 6.0$ eV to check whether insufficient localisation was responsible for the outcome.

Configuration	ΔE per Ni, $U = 4$ eV (meV)	ΔE per Ni, $U = 6$ eV (meV)	Ni moment (μ_B), 4 / 6 eV
FM	0 (reference)	0 (reference)	1.101 / 1.297
A-type	+79	+98	1.046 / 1.269
C-type	+171	+204	0.927 / 1.231
G-type	+194	+268	0.647 / 1.036

$U_{\text{eff}} = 4.0$ eV or 6.0 eV applied to the **Ni 3d shell only**.

Magnetic ordering at $U = 4$ and $U = 6$ eV



FM is lowest at both values of U ; raising U increases rather than reduces the separation from the antiferromagnetic states, closing off further U tuning as a remedy. So, we carry on using PBE+U (4eV)

Structural relaxation

Parameter	Experimental	PBE+ U relaxed	Δ (%)
a (Å)	5.2358	5.2316	−0.08
b (Å)	5.4932	5.5819	+1.62
c (Å)	7.4747	7.4851	+0.14
V (Å ³)	214.98	218.58	+1.68

Predicted lattice parameters for comparison against XRD. The volume overestimate of 1.7% is typical of PBE+ U ; the expansion is anisotropic, falling almost entirely on b .

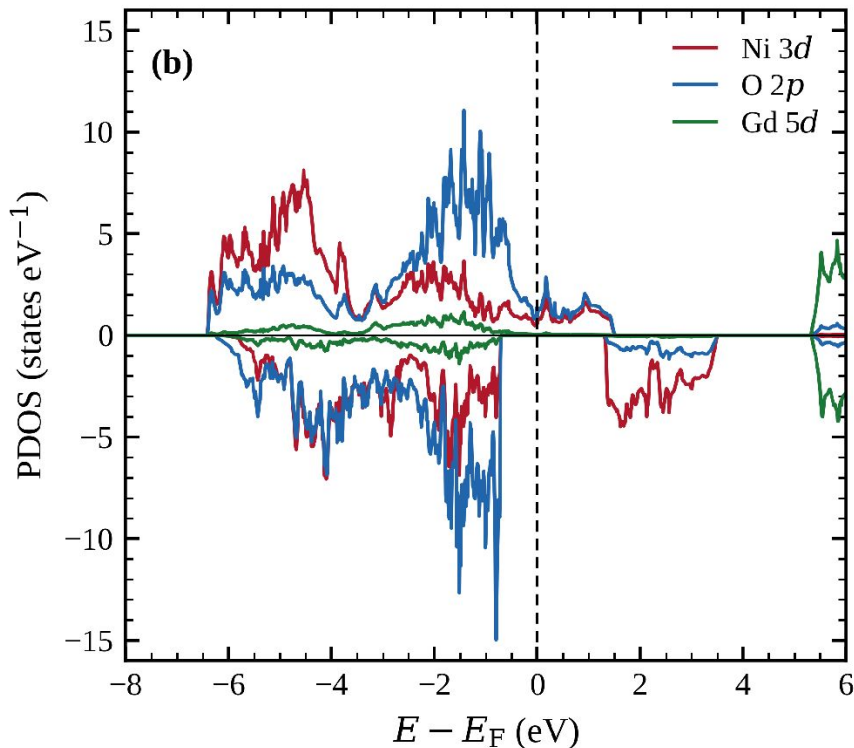
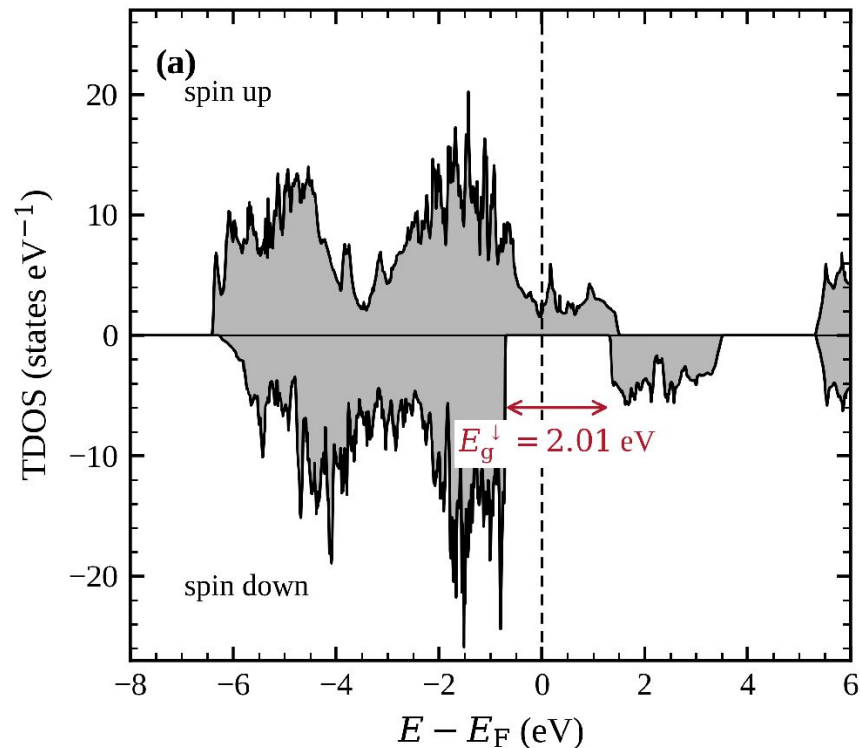
Full cell and ionic relaxation (ISIF = 3, IBRION = 2, EDIFFG = -0.01 eV Å^{−1}), FM configuration. Converged in 19 ionic steps with a residual external pressure of 0.01 kbar, indicating negligible Pulay stress and removing the need for a restart cycle.

The relaxed NiO₆ octahedra have Ni–O bond lengths of 1.9391 Å (×2), 1.9416 Å (×2) and 2.0289 Å (×2), mean 1.9699 Å, with Ni–O–Ni angles of 149.6° and 148.9°. The 0.09 Å spread is octahedral tilting; the long/short *sublattice* splitting characteristic of the breathing distortion is identically zero, as required by the $Pbnm$ symmetry.

Static self-consistent calculation

Quantity	Value
Total energy (eV/cell)	-136.36587
Basis-set (Pulay) correction vs relaxation (meV)	-8.56
Fermi energy, E_F (eV)	4.4281
Total valence electrons	147.99999 (nominal 148)
Ni magnetic moment (μ_B)	1.110
Total cell magnetisation (μ_B)	4.00009

Electronic density of states



Using denser $9 \times 9 \times 7$ k-mesh using the tetrahedron method with Blöchl corrections (ISMEAR = -5), NEDOS = 3001.

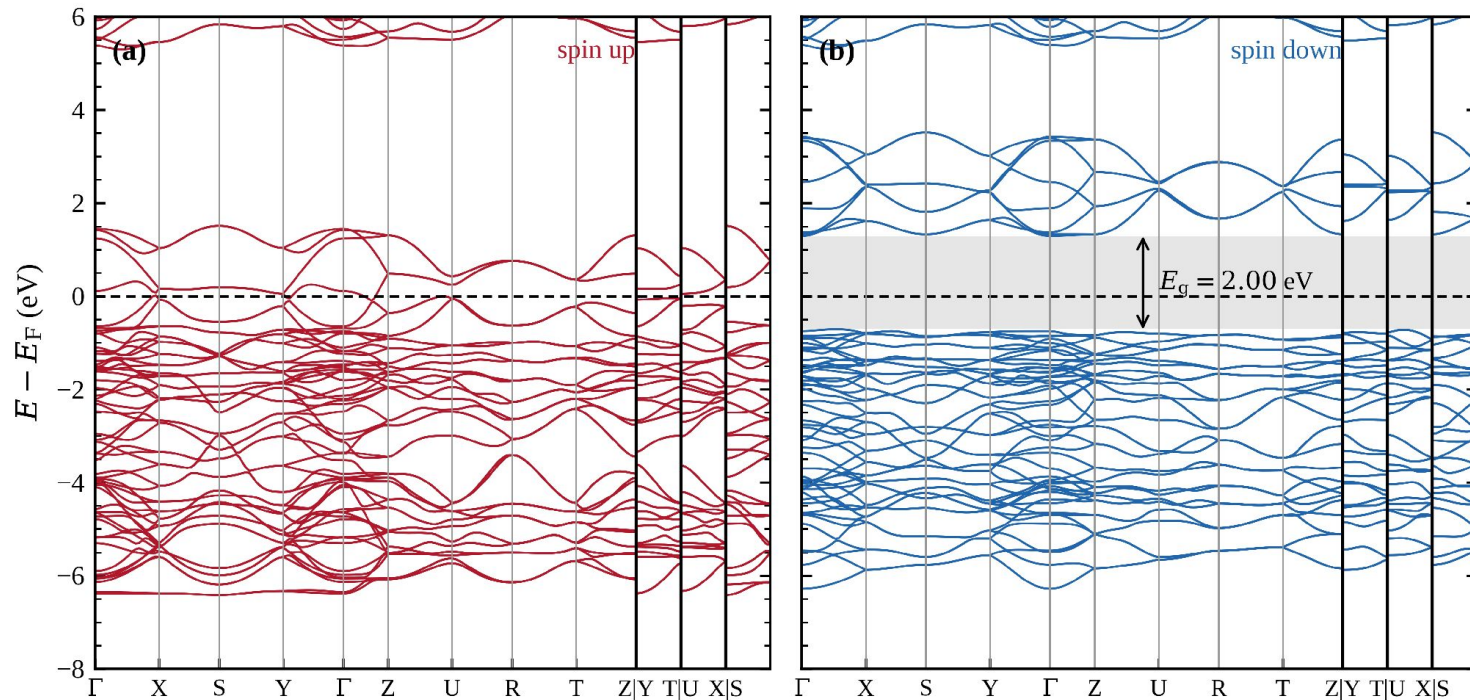
Electronic density of states

Quantity	Spin up ($\uparrow$)	Spin down ($\downarrow$)
DOS at E_F (states eV^{-1})	1.836	0.000
Band gap, E_g (eV)	0 (metallic)	2.01
Character	Metallic	Insulating

Within PBE+U, GdNiO_3 is a ferromagnetic half-metal: one spin channel conducts, the other has a 2.01 eV gap.

Of the states at the Fermi level, **58% are O 2p** and **38% Ni 3d**, so conduction is carried by hybridised oxygen–nickel states rather than by nickel alone $\rightarrow$ more evidence in oxygen vacancy and bader charge analysis.

Band structure



Non-self-consistent line-mode calculation from the same static charge density, along the standard orthorhombic path Γ -X-S-Y- Γ -Z-U-R-T-Z, Y-T, U-X, S-R (240 k -points, 120 bands).

Band structure

Only two bands cross the Fermi level, both in the spin-up channel, with crossings near X, along Y- Γ and along Γ -Z. The resulting Fermi surface is small, consistent with the modest density of states of 1.836 states eV⁻¹. The spin-down band edges lie at -0.712 eV and +1.292 eV, reproducing the density-of-states values to within one energy grid spacing and providing an independent confirmation of the half-metallic assignment on a different k -sampling.

A metallic spin-up channel means bulk electronic conduction remains available, supporting charge transport through the active material.

Formation energy

Reaction formula

$\frac{1}{2} \text{Gd}_2\text{O}_3 + \text{NiO} + \frac{1}{4} \text{O}_2 \rightarrow \text{GdNiO}_3$ (Because GGA overbinds O₂, and our formula involves only $\frac{1}{2}$ oxygen atom)

$\Delta E = E(\text{GdNiO}_3) - \frac{1}{2}E(\text{Gd}_2\text{O}_3) - E(\text{NiO}) - \frac{1}{4}E(\text{O}_2) \rightarrow \text{Formation energy}$

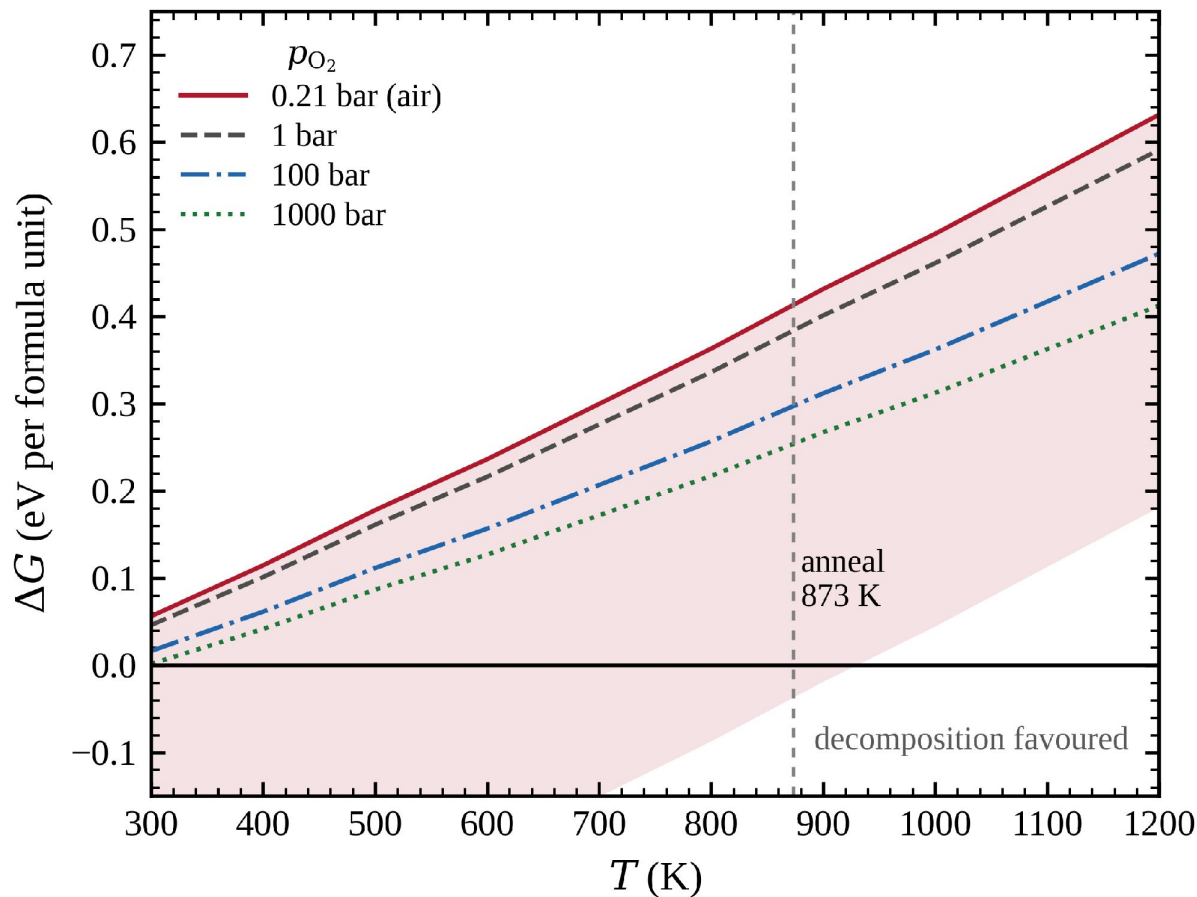
Computational consistency

All four phases share ENCUT = 600 eV, PREC = Accurate, LASPH = .TRUE., LMAXMIX = 4, EDIFF = 10^{-6} eV, EDIFFG = $-0.01 \text{ eV } \text{\AA}^{-1}$, and POTCARs.

Relax $\rightarrow$ Static

Phase	Cell	<i>k</i> -mesh	Magnetic setting
GdNiO ₃	<i>Pbnm</i> , 20 atoms	$6 \times 6 \times 4$	FM, Ni
Gd ₂ O ₃	<i>Ia-3</i> primitive, 40 atoms	$4 \times 4 \times 4$	Non-spin-polarised
NiO	AFM-II magnetic cell, 4 atoms	$7 \times 12 \times 12$	AFM-II
O ₂	$12 \times 12.5 \times 13 \text{ \AA}$ box	Γ only	Triplet

Reaction free energy against temperature



Reaction free energy against temperature at four oxygen pressures. Positive ΔG favours decomposition to $\text{Gd}_2\text{O}_3 + \text{NiO}$. The shaded band indicates the estimated systematic uncertainty of the DFT+ U reaction energy (≈ 0.45 eV).

$\Delta G < 0 \rightarrow$ formation spontaneous $\rightarrow$ perovskite stable

$\Delta G > 0 \rightarrow$ formation not spontaneous $\rightarrow$ the binaries win

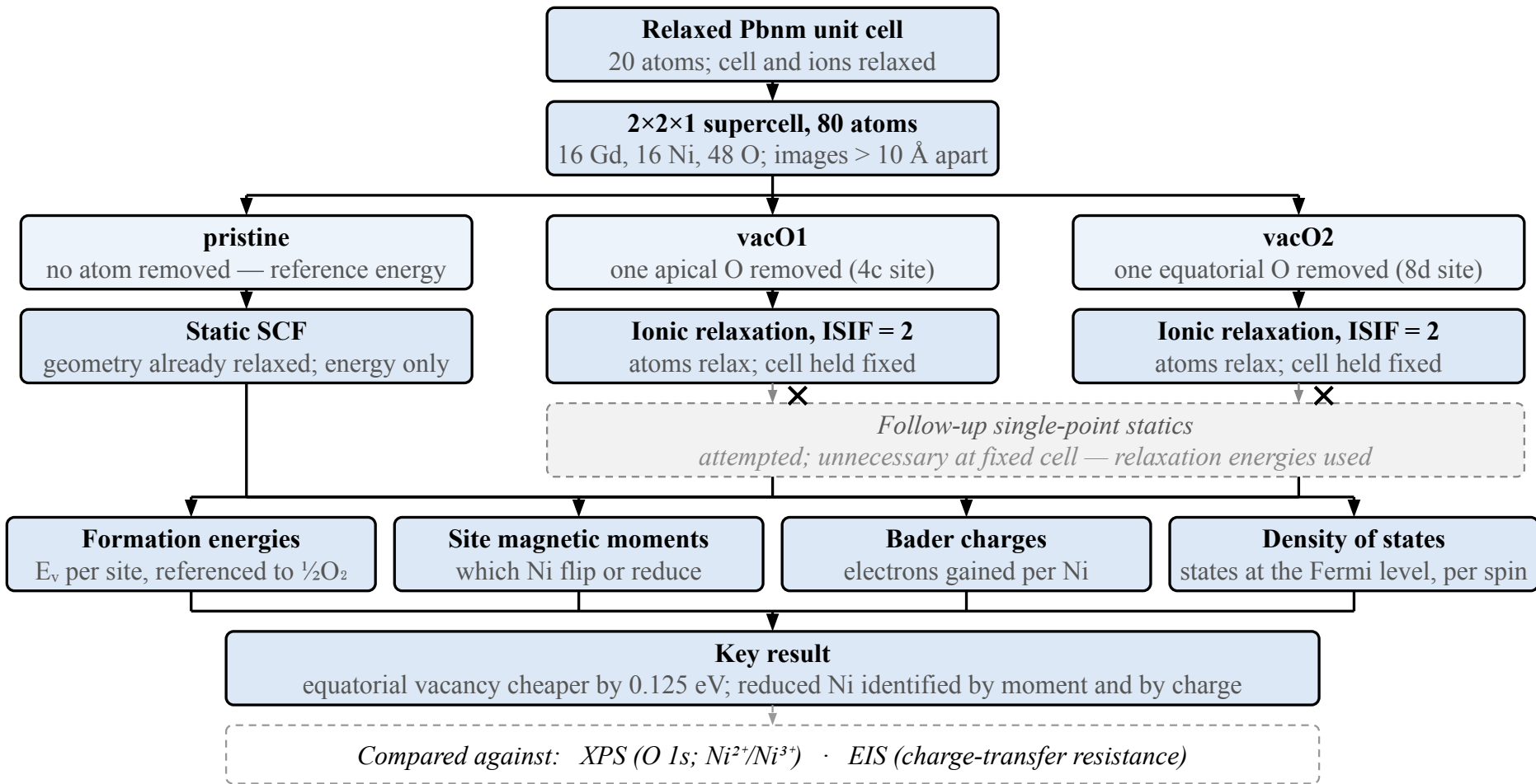
Formation energy

Phase	Cell	E (eV per f.u.)	Validation
GdNiO ₃	<i>Pbnm</i> , 20 atoms (4 f.u.)	-34.0915	Relaxed, FM, half-metallic
Gd ₂ O ₃	<i>Ia-3</i> primitive, 40 atoms (8 f.u.)	-41.7805	$a = 10.807 \text{ \AA}$ (exp. 10.813)
NiO	AFM-II cell, 4 atoms (2 f.u.)	-10.6399	Ni moments $\pm 1.643 \mu_B$
O ₂	$12 \times 12.5 \times 13 \text{ \AA}$ box	-9.8903	$d = 1.233 \text{ \AA}$ (exp. 1.208)

Applying, $\Delta E = E(\text{GdNiO}_3) - \frac{1}{2}E(\text{Gd}_2\text{O}_3) - E(\text{NiO}) - \frac{1}{4}E(\text{O}_2)$

$$\Delta E = -0.089 \text{ eV per formula unit} \\ (-8.6 \text{ kJ mol}^{-1})$$

Oxygen-vacancy workflow: GdNiO_3 , $2 \times 2 \times 1$ supercell



Cost of creating Oxygen Vacancies

Vacancy type	Site	Total energy (eV)	Formation energy (eV)
Apical (O1)	4c	-538.73340	1.790
Equatorial (O2)	8d	-538.85795	1.665
Difference	—	—	0.125

Both values fall between 1.5 and 2.5 eV, the range typical of reducible perovskite oxides. This is low enough that oxygen vacancies should form readily under normal processing, which is favourable for an electrode material.

The equatorial oxygen is easier to remove, by 0.125 eV, with an uncertainty of roughly 0.7-0.8 eV.

Where the released electrons go: Bader charge analysis

Cell	Ni site	Electrons gained	Moment (μB)
Intact	all 16 (identical)	—	+1.110
Apical vacancy	two neighbours	+0.098 each	−0.700
	most others	+0.01 to +0.03	+1.14 to +1.37
Equatorial vacancy	most reduced	+0.137	−1.606
	Second most reduced	+0.105	−0.659
	others	−0.04 to +0.05	+0.42 to +1.55

Most of the released charge does not go onto nickel at all. Reduction in this material is a shared, oxygen-mediated process rather than a clean transfer of two electrons onto two nickel ions.

Effect on electrical conduction

Cell	Majority spin DOS	Minority spin DOS	Character
Intact	0.54	0.00	half-metal
Apical vacancy	0.61	0.13	both channels conduct
Equatorial vacancy	0.38	0.00	reduced half-metal

The two vacancies again act in opposite directions. **The apical vacancy closes the minority-spin gap and raises the total density of states at the Fermi level**, so both spin channels become available for conduction. **The equatorial vacancy preserves the half-metallic character** but reduces the majority-spin density of states by about 30%.

How DFT connects to Real Measurements (XPS, EIS)

Real Measurement	DFT Result that addresses it
XPS, O 1s vacancy shoulder	Formation energies of 1.67–1.79 eV
XPS, Ni ²⁺ /Ni ³⁺ ratio	Reduced Ni is identified by moment and by charge in DFT
EIS, charge-transfer resistance	Density of states at the Fermi level

Title

Title